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Exploring the complexity of participation

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This research aims to re-articulate the concept of participation in design research by adopting a relational, temporal and process-oriented perspective. In the paper we draw on recent theory where human interaction is understood as complex responsive processes of relating. Thus, participation is seen as people taking part in creating social objects that emerge in the complex social processes of everyday organisational life. This perspective challenges conceptions of participation as normative as well as the tendency to make methods of participation generalizable and replicable. Based on an organisational case in which multiple stakeholders are involved in collaborative design processes, we describe how social dynamics and issues of power, enable and constrain participation and argue that participation cannot, strictly, be accounted for on the foundation of designed temporary workshop spaces, methods and tools for engagement or the facilitation of the researcher or designer. On that basis, we will introduce a concept which argues that participation is how people engage the temporal emergence of daily interactions and on-going conversations that unfold in complex processes of relating, influencing action, interaction, identity and meaning.

Keywords: Participation; complex responsive processes of relating; co-design; infrastructuring; participatory design; interdependency

1. Introduction

A range of research fields and industry practices have adopted an idea of participation as a way to involve end-users and publics under different umbrellas, such as participatory design, user-centred design, co-design and design thinking, each with a unique history and development. As researchers and practitioners alike have sought to engage a wider range of stakeholders in a variety of design research practices, the notion of participation takes on new meaning in accordance with the focus of the particular theory and/or practice of engagement. In current design research, we find that designers

and researchers often take assumptions of the idea of participation for granted (Winschiers-Theophilus 2012; Saad-Sulonen et al. 2018) leaving much room for individual interpretation, as few contributions articulate what participation actually might mean. This calls for a more thorough reflection on current understanding of participation, the challenges and opportunities these engender, and possible directions for the future.

Our paper sets out to reflect on and re-articulate the complexity of participation in design research practice. We will argue for an understanding of participation as the on-going interaction between those involved in emergent processes of relating. Although this perspective might seem straight-forward, it challenges normative and pre-defined understandings of participation. Our perspective originates from more recent organisational research in which human interaction is articulated as *complex responsive processes of relating* (Stacey et al. 2000). This theory focuses on the temporal emergence of social dynamics, including power relations. We draw parallels to the current discussions in some strands of design research, such as participatory design (PD) and infrastructuring, focusing on how we can understand the nature of participation, when taking into account the non-linear, fluctuating and spontaneous character of human interaction.

Looking into participatory design research, and co-design circles in particular, participation has increasingly become a focus of attention (Halskov and Hansen 2015). Originally, participatory design emerged as an ethical statement meant to be done by, for and with people (Greenbaum and Loi 2012). It sought to equalise power relations at a time when political conflicts were perceived as the main catalyst for change. Participatory design has historically focused on encouraging ideals of equality; beginning in workplaces with a commitment to worker participation and an effort to rebalance power ratios between users and technical experts, and between workers and managers (Kensing and Blomberg 1998; Greenbaum and Loi 2012) with the hope of creating better workplaces for employees. From such an emancipatory perspective, participation can be understood as an ideological and ethical demand for researchers and designers to engage particular groups. With that in mind, Kyng (1998) recognises that the design process is ‘a political one that includes conflicts at almost every step of the way’ (ibid 1998:23). While PD emerged with a strong political focus, this motivation has receded in recent design research to focus more on methods and processes of participation (Light and Akama 2012).

By considering recent CoDesign contributions, we have identified a pattern of three recurrent themes of central concern regarding the concept of participation:

1. Over the years, there has been a tendency to fluctuate between associating participation with democracy, emancipation and social power structures (Nygaard and Bergo 1975; Bjerknes et al. 1987; Bjerknes and Bratteteig 1995; Greenbaum and Loi 2012), and

seeing participation as a way of inviting users into design processes. Common for both perspectives is the insistence on developing a sense of empowerment and competence, and the ideal that collaborative, democratic relationships can help better the world and its social ecologies. While PD for a long while developed a focus on inviting users to communicate, collaborate and commit to shared goals of design inquiry, in recent years, we notice a renewed focus on the field's political attention (Basballe et al. 2016; Dixon 2018; Bønnelycke et al. 2018; Agid and Chin 2019). Dixon (2018), for example, is concerned that the commitment to democracy is abandoned and argues that a return to a pragmatist understanding of democracy could advance a transformative understanding of participation with wider political implications for PD.

2. Some strands of design research focus on participation as a process that needs to be staged (Brandt et al. 2008) through planned events or co-design workshops. For some researchers, this leads to a focus on workshops and infrastructures designed as dedicated spaces for participation where exploration and transformation take place (Simeone 2018; Salmi and Mattelmäki 2019). In contrast, other contributions challenge the notion that participation is limited to single events and recognise the need to understand participation from a process perspective (Halskov and Hansen 2015; Saad-Sulonen et al. 2018). While some contributions in the PD community address this concern by exploring the temporality of participation (Bratteteig and Wagner 2014; Saad-Sulonen et al. 2018), we notice that much research takes its point of origin in the idea of staging and scaffolding as central to participation.

3. Facilitation is, within design research, generally perceived as pivotal for participatory activities to happen. The main argument appears to be that facilitators mediate participation through methods and tools to help those involved, primarily users, to communicate their experiences, beliefs and hopes for the future (Lucero et al. 2012; Sanders and Stappers 2014; Madden et al. 2014; Basballe et al. 2016; Knutz et al. 2018). Such material and immaterial methods and tools, which gradually have become central to design practices, are argued to help create fictional spaces of reflection by acting as catalysts for dialogue (Huybrechts et al. 2018). And although the role of facilitation is critically reflected (Light and Akama 2012; Aguirre et al. 2017; Saad-Sulonen et al. 2018), this leaves us with a feeling of an idealised conception of the facilitator and the methods used to facilitate. As an example, a recent CoDesign contribution by Salmi & Mattelmäki (2019) draws on complex responsive processes of relating, in a similar way to what we attempt to do in this paper. Still, they highlight facilitation as an aim *to create favourable conditions for new experiences to take place* (ibid: 15).

Agid and Akama (2018) highlight the need to view design processes as constituted and impacted by the messy, social processes we take part in. They argue that "working *with* disruption and collective reimagining emerges as a means for building collaborative capacity" (ibid: 808). They name this the *dance of designing*, which orients towards the temporality and emergence of participation. We agree with this perspective, but through their argumentation we sense that facilitation is understood as being a central part of this dance.

A similar focus on temporality has emerged in Infrastructuring. Bødker et al. (2017) frame infrastructuring as interrelated organisational, social and technical activities which develop over time through the involvement of users. The term developed from Star and Ruhleder's (1996) idea of information infrastructure. While originating in the spatial expression "infrastructure", Karasti et al. (2018) move toward a temporal language, and notice that "infrastructur*ing*" carries with it a shift from a fixed ontology to a 'process ontology'. Björgvinsson et al. (2010) assert that infrastructuring is an ongoing arrangement, which involves people, objects and processes, with the aim to ensure that science and technological developments are structured in ways that are democratically beneficial to citizens. As such, infrastructuring addresses some of the challenges related to an increasingly connected world and focuses on the politics in processes of collaboration. This aligns with other PD research, and, for example, Light and Akama (2014) who argue that when politics gains the centre of attention, then designers can contribute to a democratic society.

Bødker et al. (2017) argue that infrastructuring moves between different social infrastructures where multiple people participate over time across hierarchical levels and that infrastructuring is thereby not limited to a horizontal series of activities. Within the PD community, infrastructuring research focuses on design activities such as workshops where stakeholders collaborate on the physical manifestation of design, and on aspects of local processes of participation, such as methods used in workshops. They also argue there has been a lack of focus on the messy, 'back-stage' processes that temporarily emerge beyond specific design activities and workshops.

We find that more recent organisational research, in particular the theory of *complex responsive processes of human relating* (Stacey et al. 2000) - henceforth, written as: complex responsive processes - can contribute to design research, such as PD and Infrastructuring, by bringing forth a perspective of participation that focuses on the nature of human interaction. Similar to PD and Infrastructuring, the theory of complex responsive processes acknowledges that power relations and politics of everyday interactions are central to the nature of participation. Yet, highlighting non-linearity and thereby avoiding pre-defined ideas of staging, facilitation or democracy.

Complex responsive processes has previously been drawn upon in CoDesign contributions as means to make sense of human interaction, for example with a focus on the quality of conversations (Buur and

Larsen 2010) and more recently in two papers; one by Salmi & Mattelmäki (2019), which focuses on co-design facilitation in temporary, in-between spaces and another by Andersen & Mosleh (2020), focusing particularly on the interplay of intentions between multiple stakeholders in co-design activities. However, a complex responsive process perspective has not yet been leveraged to help us understand and more clearly articulate a relational perspective on the concept of participation.

Based on the above, we make room for an on-going conversation about what participation is and insist that we need to get specific about what we mean, when we say participation is at work. Our aim with the paper is to offer a framework for critical reflection on what participation means, when focused on human relating and to challenge inherent understandings of participation in design research literature. As such we will address the following research question: *How can the theory of complex responsive processes of relating help us understand and challenge predominant ways of understanding participation in collaborative design practice?*

In addition to reflecting on the key elements of complex responsive processes, we will describe and draw on our experience of an organisational case and use these to explore and exemplify the consequences of taking this theoretical perspective on participation. Although we limit this research to an organisational context, we suggest that our findings can help expand the notion of participation in relation to collaborative design practice in general.

1.1 Complexity science and non-linearity

In the 1990's, the insights of complexity science had a huge impact on natural science (Kauffman 1996; Prigogine 1997; Allen 1998). Large patterns such as the weather, stock rates and biodiversity were understood to emerge as a result of local, temporal processes of interaction between small entities. Prigogine (1997) argues in the book "The End of Certainty": *"What is now emerging is an "intermediate" description that lies somewhat between the two alienating images of a deterministic world and an arbitrary world of pure chance"* (ibid 1997). What Prigogine refers to is that patterns that emerge over time are often recognisable and in no way random; however, they should not to be seen as predictable.

In a paper on the Design and Cultures of Enterprises, Deserti and Rizzo (2014) argue that design culture is '*a specific system of knowledge, competencies and skills that operates in a specific context to develop new products and services'* (ibid. 26). Similarly, Lurås (2016) argues for *'a systemic view on the design situation'*, which involves the system we design in, the system we design and the system we design for, where each system has its overall goal.

Reflecting specifically on service design, Agid and Akama (2018) notice a tendency to consider design as a 'neutral' site, in which systems thinking strips away social, cultural and political

dimensions. In its purest form, this perspective tends to eliminate power relations, and ideas of temporality and context. What Agid and Akama notice, resonates with the perspectives of complex responsive processes. For Stacey, Griffin and Shaw (2000) designing technical systems in organisations differs from understanding oneself as part of a "human system". They argue that an organisation exists not as a system but emerges as a *social object* (Stacey and Mowles 2016), which those involved continuously create and recreate through their on-going interaction. George Herbert Mead (1934) described social objects as generalised tendencies among people to act in similar ways in similar situations. From this perspective, organisations become what they become through iterations of people's local, temporal interactions, which as a result maintains the idea of "the organisation" as reasonably stable. While other researchers understand non-human structures as forces with their own agency (Latour and Akrich 1992; Parmiggiani and Karasti 2018) the concept of social objects is, by Mead, described as the insistence on arguing that only humans have agency/ intentions. As such, Mead (ibid) avoids reifying the notion of an "organisation" and instead insists that social objects are created and recreated through human interaction.

1.2 Complex responsive processes of relating

When Stacey, Griffin and Shaw published their critique of systems thinking, they presented a radical understanding of social interaction as *complex responsive processes of relating*, where in on-going local *processes of relating* people *gesture and respond* to each other to create complex webs of interdependency as they engage their intentions, agendas and interests.

What continuously emerges from this gesturing and responding is therefore *complex*. The emergence of these processes and the global patterns they create are unpredictable, yet often recognisable. As such, organisational patterns emerge as a result of the interplay of many people's intentions, not as the outcome of someone's individual agenda.

Stacey et al. (2000), Stacey (2001), Griffin (2002) and Shaw (2002) draw on social psychology, in particular the work of the pragmatist philosopher George Herbert Mead, a close colleague of John Dewey, and his idea of conversation as simultaneously transforming mind, self and society (Mead, 1934), and on the process sociologist Norbert Elias and his idea of figuration and power as human interdependency that involves processes of inclusion and exclusion (Elias and Scotson 1994; Goudsblom and Mennell 1998).

Mead asserts that responding cannot be split from gesturing. As action in the present moment is spontaneously responded to, over time mind and self-emerge as part of the on-going interaction of gesture and response. Hence, a single individual cannot be seen as an entity isolated from others. As Mead explains, the individual mind is an I/me dialectic, where "me," a socialised synthesis of how

other people tend to act on one's own actions, or "generalised other", and "I" is the spontaneous response from the individual towards "me". From a complex responsive process perspective, focus is on how social interaction simultaneously transforms us and the world. This radical understanding challenges the notion of communicative action as transferring messages between individual minds. Thus, Mead's theory underscores how participation emerges through the temporal interaction between people, as opposed to a focus on the individual.

An emphasis on ongoing interaction, process and temporality can also be found in infrastructuring research. The shift from "infrastructure" to "infrastructuring" is witness to that, and the call for the 2018 PDC conference focused on temporality and the politics of participation. Karasti et al. (2018) argue that infrastructuring constantly shapes power relations. They also argue that *"despite the promises of opening up participation to knowledge sharing and production, the dynamics of participatory processes that shape infrastructuring efforts remain underspecified"*.

Although both perspectives focus on power and politics, there are significant differences. In complex responsive processes, the overall pattern of an organisation emerges from local, moment to moment interaction. Here, change, as with stability, is linked to the temporal emergence of participation. The relative stability we experience is the effect of *social control.* "Because we, as humans, have the capacity to take the attitude of the other – anticipate how the other will react – we control ourselves." (Mead, 1934). To better understand the nature of power, complex responsive processes has also drawn inspiration from Norbert Elias (1998). For Elias, *"Power is not an amulet possessed by one person and not by another; it is a structural characteristic of human relationship – of all human relationships"* (Elias, 1998: 116). Elias rejected any idea that people are independent of each other. Instead he understood power as a mutual interdependency that is constantly being negotiated in the politics of everyday life as co-created figurations of interaction that involve processes of inclusion, exclusion and emotional reactions of shame, panic and anxiety.

In this regard, how we as people interact with and embrace the dynamics of participation will inevitably be bound up with ethics. Griffin (2002), reflecting on the nature of ethics, notes that participation is the direct interaction of people with each other and as a consequence what each of us does, matters. Griffin, drawing on Mead, argues that we cannot escape the responsibility of our own action, which we paradoxically cannot predict the outcome of. However, it does give us the freedom to relate to the moment to moment, emergent situation and contribute to its next iteration. This differs significantly from an understanding of ethics as universal principles to be applied.

2. Participation in an Organisational Context

Over the course of 14 months, starting April 2016, we conducted participant observation and semi-structured interviews (Blomberg et al. 1993) with three department managers and eight practitioners who worked in the technology development department of a large European manufacturing company. To better understand their innovation practice, we regularly visited the company and closely followed the department's employees in project meetings, strategic management meetings and innovation training workshops. The research collaboration was initiated through a project focused on developing new methods for the training of practitioners in innovation and entrepreneurship processes. The project involved six industrial partners and four academic institutions, where each academic group was paired with a company to explore their particular innovation challenges and develop one of eight methods for that challenge. The case company we describe became involved via a previous collaboration with the Head of Department. The stakeholders we involved in the research process were gathered as a task force based on his initiative.

As we engaged the participants, in both formal - meetings, etc. - and informal - chance encounters etc., it became clear that in this engaging, the informal conversations began to play a more significant role than our formal research methods. In fact, we became aware that our presence altogether began to align with Stacey and Griffin's (2005) notion of researching as a relating to the experience of the present moment. The narratives that emerged as a result enabled us to recognise new patterns and themes that we invited the stakeholders to engage.

It should be noted that the structure of this section (2.), reflects the temporal order of events as they occurred. We do this, as we are interested in engaging the reader in how the participation of those involved influenced the dynamics of the situations that arose.

The organisational department we studied is divided into sub-sections of design, mechanics and electronics, and employs designers and engineers. The department recently went through a shift in development practices due to strategic, organisational changes. Prior to the change, the responsibility of engaging end-users was in the design sub-section, where those concerned handed over insights and new product concepts to the engineering sub-section. A new department manager was hired, and part of his responsibility was to 'break down silos' and require employees to collaborate across disciplines. Both engineers and designers were expected to engage each other's field of expertise to ensure that no resources were wasted by, for example, making design changes due to a lack of technical feasibility.

In order to streamline these decisions, two other changes were introduced: The Standard Committee and The Component Portfolio. Six senior engineering experts and one designer were elected by management to be part of a standard committee established to limit the technology component portfolio and ensure that new designs either used the company's existing component portfolio or that

any new component could have a wider use, thus minimising development and manufacturing resources.

2.1 Participating in Conversations Across Disciplines

Some of the engineers opposed the cross-disciplinary work claiming that management was neglecting their professional interests. However, other, recently employed engineers, collaborated more willingly with the designers, which raised a number of tensions. In order to improve the innovation process and create a willingness amongst the employees to work together, management attempted to facilitate conversations between the two professional groups. As a result, new collaboration guidelines were presented to the group.

However, the senior experts in the standard committee found it difficult to participate in cross-disciplinary conversations and decision-making processes designed to streamline the component portfolio. This is reflected in a response from one of the mechanical engineers: *"We might as well face it. We don't understand each other's priorities. Sometimes it is my feeling that engineers can have problems understanding why some things are so important for the designers … LEDs are just components that make light, and whether the colour is pure green or dark green doesn't matter to us. Of course, some things are more serious than that. There are cases where we immediately just have to say no. If it is not technically feasible, then we might as well kill the idea".*

By participating in these conversations, the senior experts form and are formed by each other. Their skills and interests are continuously unfolding in their recognition of how they act and how others act. As Stacey (2003) explains, collective and individual identities are formed in local interactions and are reflected in the experience of being together as they provoke, agree and disagree. Conversations emerge as a negotiation of patterns, not just in actions or words, but in the responses they offer. Participation is not to be understood here as the simple adoption of the other's point of view, but as an engagement in the on-going negotiations without knowing whether and how the conflicts transform into new conversation or just deepen stuck patterns and conflicts.

2.2 Participating in the Standard Committee

Due to the structural changes, employees experienced a misalignment in levels of experience with and sensitivity to the existing product portfolio. In our fieldwork we saw some members of the standard committee becoming very frustrated due to their not knowing what was expected of them. They experienced the committee meetings as highly unproductive and found it impossible to make the requested decisions.

It appeared that the standard committee engineers were caught between two changes made in the department; on the one hand they were expected to engage the designers in exploratory concept development and on the other, they were supposed to restrict the introduction of new components. This created a pattern where they were perceived as spoilsports when they acted on their committee brief. An experienced engineer pointed out: *"I guess they (the designers) look at us and think that the engineers are the ones always saying no. We are also learning that we are not allowed to say no. But the designers also need to learn, that maybe they should listen to us. That maybe we are right!".*

Altogether, we noticed how the participants in their on-going interaction with each other and through their inclusion and exclusion of each other, were co-creating meaning; not as nicely organised viewpoints, but as interaction figurations (Elias 1998) that emerged from their participation.

2.3 Participating in Emergent Power Relations

Gradually, after months of formal and informal conversations, we could see how those involved negotiated alternative ways to collaborate, which in turn influenced their sense of professional identity. Then, what was called "the organisational reality", stepped in. With a large number of projects looming, time became the primary constraint. To keep sales up, technical feasibility and ease of manufacturing were prioritised, which did limit the freedom for new designs, user studies and testing. As such, the designers had to compromise their interests, which influenced their professional identity. The pressure to deliver demanded they better understand the technical restrictions and discover ways to incorporate their expertise. This led to their finding alternatives as to how to negotiate their priorities. One of the designers we interviewed said: *"Production always wins. …(but) to be honest, we as designers are finding our own ways. Instead of negotiating too much with the engineers and our managers, we build other relations in the organisation to negotiate the importance of our work".* So, what was articulated as "the organisational reality" became another theme for local negotiation, with the designers initially feeling their expertise was seen as less important. However, feeling constrained by the engineers, this forced them to resort to alternative ways of negotiating their roles and participation in the development process. They realigned their web of interdependencies and by doing so, became less dependent on the engineers and managers. The term "the organisational reality" is not an objective fact in itself, rather a co-constructed understanding that emerged as a *social object* (Stacey and Mowles 2016).

Referencing Elias (1998), Stacey and Mowles (2016) argue that interdependency is central to understanding the nature of power, where the mutual dependency of relations between people can also be understood as power relations. Thus, power is not one person exerting power over the other, or as Elias puts it: "an amulet one can have". When people simultaneously constrain and enable each other in processes of inclusion and exclusion, it influences the identity of those involved. Equally, the re-

negotiations of the social dynamics between groups of people and the conflicts or disruptions that arise, most often lead to the emergence of new patterns of meaning.

This was evident in how the designers and engineers opposed increased collaboration, how the younger engineers negotiated their own identity by distancing themselves from the more experienced and how the designers developed alternative relationships to wield their influence. Thus, to participate in the department meant, in practice, to participate in maintaining and disrupting existing patterns and forming new ones as a response to each other and to the procedures management proposed.

2.4 Participating in Managerial Issues

The managers of the sub-sections also participated in conversations that challenged their roles and interactions with each other. Here, the manager of the mechanics sub-section explains how he felt his management colleague in the design sub-section manipulated his own senior design experts to enforce a certain agenda in the standard committee. *"He sees them (the committee) as something he can put in a box and ask to develop a standard component. The problem appears when projects need much less time to finish than a standard component that fulfils all requirements. This means that the projects then want specially designed components on time. So, instead of focusing on making standardised components, the committee needs to deliver what the projects want. And instead of having one component for all projects they have 114. That brings us back to ground zero".* He proposed having a chief technical component specialist on the committee and we got the feeling that he wanted to be that chief specialist. So, while the employees had their difficulties with the managers, they had their own disagreements and conflicts that influenced the patterns that emerged.

Stacey (2001) highlights the interplay of intentions among stakeholders. He argues that nobody can control what others do, nor the outcome. This creates uncertainty, which becomes the lived reality of everyone involved. As a consequence, all participants have to deal with the potential for surprise and the paradox of knowing and not knowing. As researchers, we saw how conflicts emerged from the crossing intentions and interests of the stakeholders and affected the local transactions in the department which, in turn, opened up for negotiations about the development process.

2.5 Participating in the relating of an improvised theatre workshop

Faced with the challenges described above, we organised a series of workshops where tangible artefacts (Mosleh 2017; Andersen and Mosleh 2020) and improvisational theatre were used (Larsen 2006; 2011).

One workshop was organised to accommodate the wish of the two managers to reflect on the role of the standard committee. At the time we experienced a range of tensions, between: the managers and

committee members, the two managers, members of the committee itself, and members of the committee and other engineers.

From experience, we know that using improvised theatre is a way of inviting stakeholders to engender conversations about themes they struggle with. In previous literature we have shown how, by enacting a fictitious, yet relevant scenario on stage, this allows participants to engage those themes, as we never directly address conflicts between individuals.

Present at the workshop were the two managers, the members of the committee and about ten engineers from the department. We (the two authors) took part, together with two professional actors who had many years of experience of working with improvised theatre in organisations. Over the past months, one of the authors had been involved in a research collaboration with all the participants present. Furthermore, inspired by previous interactions with the actors, some were already using improvised theatre in facilitating their own group work.

We asked the standard committee members to act as directors for the enactment of the "standard committee" meeting that was to be staged. Together with our two professional actors we had one of the managers and three engineers from the department on stage.

The two professional actors started by playing a small, rehearsed scene between an engineer and a designer. The scene described a decision that needed to be made in collaboration with the standard committee members and, as such, related to the performers' experience of working across disciplines and management's involvement.

So, directed by the real committee members, a fictitious meeting revealed tensions and conflicts to the audience that were both recognizable and relevant to the group. One of our actors introduced a dilemma as part of the improvisation: A designer contacted an engineer to see if he could use a particular material in his design. The engineer told the designer that he would have to ask the committee's advice as to what to do before he could answer. At this point, the facilitator of the workshop stops the scene and turns to the real committee members directing the play, to hear what they would do. The committee's response was prompt: There would be neither the time nor precedent for that kind of discussion in the committee.

As the play continued, it quickly became clear to the audience that this lack of a decision would impact the engineer's future relations with the designer. One of the engineers playing a committee member was newly employed but had years of experience collaborating with the company. He took the opportunity to challenge the committee's approach, by asking them to forget what they did in the past: *"Listen, things have changed, we cannot wrap this place around what happened before, so let's hear what they have to say"*.

Although, at the time, we as researchers did not know this new employee was going to be part of the committee in the future, it became obvious that his participation in the enacted, fictitious meeting impacted future meetings in the real committee.

As the conflict heated up, we asked the audience to imagine how the situation could unfold. Although a fictitious situation, their response reflected their experience of similar situations. This led to several people asking both managers to be on stage. The managers realised they would lose face if they rejected the invitation. On stage the managers argued that the experienced engineers, as members of the standard committee, had a better background for making decisions on a daily basis. Gradually the situation shifted from working solely with improvised theatre to discussing the themes that the participants were in reality struggling with.

By the end of the session, all agreed they had taken part in an important discussion that had not previously surfaced. The managers realised they should engage their staff further and members of the audience who were not members of the committee gained insights into the dilemmas the committee experienced. Some said this made them realise they also had a shared responsibility for getting things done. In other words, the quality of their conversation changed, and their relational understanding changed. That said, we do not claim there was a shared goal or mutual agreement reached between the participants; we simply noticed a shift in the otherwise stuck pattern of conversation.

As researchers, we would not have been able to stage the standard committee's specific challenges without having participated in the formal and informal conversations in the company over a longer period of time. Furthermore, we had to live with the risk of not knowing the outcome when we invited the stakeholders to participate. Reflecting on our preparation and contribution to the workshop, it is clear that it was based on purposeful planning (Simeone 2018; Salmi and Mattelmäki 2019). However, what we found important was our invitation to engage "quality of conversations" (Buur and Larsen 2010). In the 'staging' we deliberately initiated an improvised conversation with the standard committee who directed the enacted meeting, which was influenced by how the actors played out the meeting and which was further influenced by comments from the audience. Using improvised theatre allows people to more freely engage with themes of conflict, since the fictitious nature of the work allows for risk-taking (Larsen 2011). We were fully aware that what emerged could be challenging for the two managers, as we couldn't know how they might interpret the situation or our role. As researchers, we had to run the risk of not knowing how the participants would react and improvise along the way.

This aligns with Salmi and Mattelmäki (2019) who describe workshops as temporary, co-habited learning platforms that relate to dynamics that lie beyond a workshop, for example, the forces at play between managers and employees. However, from our research perspective and the local company

politics we were working with, this demanded that we, in both the improvised theatre and the workshop as a whole, took the risk to navigate from moment to moment and invite a possible next move without advocating any particular group or person. So, in our "letting go," we attempted to engender moment to moment, temporal and relational moves to improvise with the situations people were grappling with, the outcome of which we couldn't know beforehand. In another workshop in the same organisation, described in Mosleh (2017), the researcher introduced the use of tangible design artefacts. The situation took a surprising turn, as the employees started using the tangible artefacts a means to raise new debates about their own, rather than management's, agenda and interests. In that case, the researcher had to also improvise without knowing what would happen. Thus, in both cases, regardless of the method being used - theatre or tangible design artefacts - we couldn't predict the outcome of the process, but instead used them as invitations to engender new conversations between those involved.

Asking people to engage in the workshops we describe above is inherently risky for all involved, as could be seen when the two managers were put on stage. Inviting such spontaneity challenges the power relations that emerge from the mutual interaction of people's everyday situations and the conflicts, tensions, paradoxes and dilemmas they experience (Larsen 2006). This invites for ethical reflections.

As mentioned above, Griffin (2002) argued that an ethical interpretation of action is found in action itself, in the on-going recognition of actions that could not be anticipated in advance. Drawing on Mead, Griffin argues that we cannot escape the responsibility of our own action, which we paradoxically cannot predict the outcome of. Although this perspective encourages us to pay attention to what we do and heed its consequences, our freedom lies in our ability to notice the moment-to-moment of an emergent situation and contribute to its next iteration, as was the case in the theatre workshop. From that, we argue that participation involves moment-to-moment negotiations that emerge as a result of the unpredictability of a response to one's own action. This differs from a grand ethics where one decides on ethical and unethical action upfront.

2.6 Participating in the creation of social objects

As mentioned earlier, Stacey and Mowles (2016) build on the notion of social objects, which was originally coined by George Herbert Mead. They do that as a way to understand what is otherwise explained as "organisations", "structures" etc. Looking at the case material, we understand the standard committee as a social object of which the meaning continuously emerges. In the work with improvised theatre, we noticed that the participants began discussing and changing the meaning of the standard committee. This negotiation involved not only the committee, but also the other employees, such as the newly employed engineer and the department managers. Over time, the meaning, e.g. the social

object of the standard committee will be negotiated and renegotiated in every small interaction that emerges between the stakeholders involved. Viewing the organisation from this perspective presents an alternative understanding of structure and on the interplay between structure and people, by arguing that we as humans negotiate and change structure through the way we create and recreate the meaning as a social object in our interactions.

3. Re-articulating Participation

A complex responsive processes of relating perspective reveals that human interaction is "*paradoxical…* (with its) *cooperation and competition/conflict, difference and sameness. However, central to the approach is the claim that the idealisation cannot be taken on its own as a given or universal because it only arises in its particularisation in contingent, local situations. (Stacey and Griffin 2005: 34).*

For Stacey and his colleagues, participation is how people engage the temporal emergence of daily interactions and on-going conversations that unfold in complex processes of relating. In the organisational case described above, we have seen how patterns of interaction emerge in what Allen (1998) named self-organised behaviour. The employees engage in conversations with different intentions and in their interaction co-construct their next moves, which for some of them is perceived as "progress" and for others a constraint. This engagement with the process in no way aligns with an idealised or emancipatory notion of participation.

The participation between those involved is characterised by inclusion and exclusion, such as the designers' attempt to circumvent the engineers and the managers' attempts to out-manoeuvre each other. The interactions between the stakeholders continuously change the interdependencies of the social figuration. Participation is improvised as stakeholders temporally act on gestures and responses from each other. None of them – nor we - can predict or control those interactions, which may evoke feelings of risk, due to possible exclusion from the social process of engagement. We can see how the managers' initiatives are met with a response, the outcome of which becomes something different. Although fixed conflicts might emerge, positions will shift, which however rarely leads to a harmonious or mutual agreement (Buur and Larsen 2010).

Complex responsive processes of relating challenges the assumption that participation in a design or development process equalises power relations or enables consensus. As researcher, facilitator or designer, one might have intentions to include stakeholders in particular ways, but because of the non-linearity of social processes, what actually happens, and how one understands the impact of one's contribution, cannot be predicted beforehand. For that reason, we suggest that we re-articulate

participation as an understanding where *each of us takes part with each our intentions and through that we create and recreate our own and the role of others in the processes of relating.* In the process, we might have intentions of including or excluding others, but we cannot be solely responsible for the outcome and we have to be aware that our actions might have unintended consequences that cannot be predicted beforehand.

3.1 Workshops as Moments of Relating

In the case above, we see how the theatre workshop resonated with the participants, as it reflected a theme relevant to both managers and employees. The theatre workshop invited the participants to confront the challenges of the standard committee and opened a forum where those present could negotiate their development practices. We consider improvised theatre as an invitation where fiction is both composed of and met by the actual experience of the participants (Larsen 2006; 2011). An invitation for a next iteration of participatory negotiation of the current figuration to be followed by another (Shaw 2002); an on-going process of negotiation where inclusion and exclusion emerge as inevitable elements of the participation.

However, particular methods and tools, such as improvised theatre, can also act as invitations for people to explore what they might usually *not* do on a daily basis (Larsen 2006). Here, participants can be encouraged to re-articulate their situation by exploring shifts in power relations, e.g. the social dynamics of inclusion/exclusion. For this reason, we argue that workshops are co-created through the participation of all stakeholders. Thus, prior planning by a facilitator cannot guarantee prescribed intentions, but should be seen more as an invitation to engender new conversations and shift positions, including those of researchers or designers.

Although workshops provide an opportunity for researchers, designers and others to engage organisational stakeholders, we must emphasise that their facilitation means taking part in the moment to moment, improvised interactions between all, where possible outcomes are a change in the social figuration of those involved. This, rather than their being designed as a predictable process. Thus, workshops are neither laboratories that create outcomes for later implementation, nor self-contained spaces built on the assumption that participation can be staged or controlled through particular forms of engagement (Brandt, 2001; Mattelmäki and Sleeswijk Visser 2011). This aligns with Light and Akama (2012), who point out that recent PD research has focused on facilitation methods, rather than on the situated processes that are central to participation.

From a complex responsive processes perspective, social objects such as organisations or the standard committee only exist in their present form because we recreate and reshape them through our temporal interactions (Stacey and Mowles 2016). In their reflections on co-designing, influenced by their

understanding of complex responsive processes, Salmi and Mattelmäki (2019) describe their interaction as researchers in a "facilitators' space" with those from an organisation in an "organisation's space" as an intertwining and collaborating in an "in-between space". We do not see this spatial metaphor as helpful in understanding how the participating and relating emerge. Although the interactions are carried out in physical space, of which the setting has an influence, the metaphor of manoeuvring within and between spaces, detracts from the temporal nature of interaction between those involved.

The use of spatial metaphors introduces a concept of participation where the dynamic flux of temporality is exchanged with fixed positions, creating a language that makes the temporality of interaction difficult to understand.

While we see a tendency to use spatial metaphors to describe co-designing processes, some researchers, specifically within service design, have raised an awareness of the relational, processual and atmospheric qualities of collaborative endeavours. Akama (2014) encourages us to attune to 'between-ness' when designing. By paying attention to the interrelations *between* people, objects, space and time and recognising the relational aspects of participation, this blurs the boundaries of the spatial, temporal, object and subject, thereby acknowledging that co-designing does not exist within predefined boundaries. In a similar vein, Akama and Prendiville (2014) oppose seeing design processes as a string of systemized processes or methods. They argue that co-designing is a process that constantly emerges and transforms relations between people and materials, and that to understand its entanglement, we need to move beyond an object and method-oriented perspective of co-designing.

Conclusion

Participation is a central theme in a variety of design research fields. Temporary workshop settings, methods and tools of participation and facilitation figure often as topics in current literature, in particular CoDesign. This focus has yielded discussions on what participation means and how it emerges.

Although some research contributions, particularly Infrastructuring theories, have highlighted the importance of addressing the social and entangled messiness of co-designing with its related "back-stage" processes that emerge beyond concrete design activities and events, we still find there is a lack of research that engages the temporal, social, complex and power relations of participation. We argue that shifting from systemic thinking to focus on the social entanglement that participation entails, requires a fundamental acknowledgement of the complex responsive processes that emerge in interactions between people.

We will now return to the three main themes we articulated at the beginning of the paper, yet now seen through our perspective of participation based on the above.

1. From a non-linear, processual perspective, such as complex responsive processes of relating, participation is the co-creation of social objects, which reflect the tendency for people to act in similar ways in similar situations.

2. The dichotomy between democracy/emancipation and engagement of users can now be seen in a different light. We all act politically in our everyday interactions and in doing so, we constantly take part in processes of inclusion and exclusion that emerge as inevitable elements of our participation.

3. Participation cannot be controlled, and the facilitation of workshops has to be understood as present moments that are negotiated between stakeholders following from and followed by other present moments of interaction. From this perspective, design research methods in general and workshops in particular can act as powerful invitations for new conversations to emerge, yet the outcome remains unpredictable. This rather than understanding workshops and other methods as staged/designed events that reach for a prescribed outcome.

4. Although we as researchers can deploy methods and stage invitations to engender new conversations, we must still respond to the on-going and unexpected outcomes. Although we might work for and with particular groups, we are not responsible for a specific outcome, nor can we expect the use of a method or tool to lead to predefined results. Yet, we have an ethical responsibility to engage the co-created outcome. In the temporal interactions in which we participate as researchers, we cannot avoid our role as stakeholders that shape and are shaped by the events and interactions that emerge (Gottlieb et al. 2013; Heape et al. 2015; Buur and Larsen 2010). Through formal and informal conversations, and temporary workshop settings, sometimes involving design materials, we actively engage in an interaction with the stakeholders and thereby participate in the social dynamics of the situation (Larsen and Bogers 2014). When we, as researchers, initiate these invitations, we cannot stand outside. At the same time, we are, just like the other participants, negotiating our participation and thereby run the risk of being excluded.

As a final conclusion, based on the above explorations, descriptions and reflections on related design research and our organisational case study, we have shown how interdependencies, conflicts and social dynamics continuously emerge, change and affect how people enable and constrain each other

through their co-design participation. Based on this, we suggest re-articulating participation as an understanding where *each of us takes part with each our intentions and through that we create and recreate our own and the role of others in processes of relating.* As such, we suggest that by taking seriously the communicative interactions that emerge between multiple stakeholders in collaborative design practice, this allows for a more nuanced understanding and re-articulation of the transformative nature of human relating and the complexity of participation to emerge.

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